Instructions

Input Cells: Cells that require user input have an orange background.

Status Indicators: Cells, sections, and worksheets each have an overall status, indicated by blue background cells.

Images and Schematics: For sections requiring images or schematics, paste the image over the specified area. Use high-resolution images to allow for zooming if needed, and then resize the image appropriately.

Datasheets: Provide a hyperlink to the manufacturer's datasheet where requested.

Additional Comments: Sections labeled "Additional Comments" are provided for documenting anything the team feels is not adequately covered in the specified fields. Completing these sections is optional.

Status

Datasheets	OK
TS Schematics	OK
Accumulator	OK
Precharge/Discharge	OK
Charging	OK
Shutdown Circuit	BLANK
Torque Security	OK
Other	BLANK

Team Name	NU Racing
University	University of Newcastle
Competition Year	2025

Max TS Voltage	453.6	V
Accumulator Fuse	100	A

Additional Comments

Template Revision History

Version	Date	Comment
2.4	9/30/24	Updated for 2025 rules and terminology, general bug fixes.
2.3	10/29/23	Updated for 2024 season.
2.2	11/3/22	Updated for 2023 season.
2.1	10/22/21	Updated for 2022 season.
2.0	11/27/20	Multiple updates and bug fixes.

Status	OK

Battery Cell		OK
Make	Sony Murata	
Model	VTC6	
Nominal Capacity	3.1	Ahr
Maximum Voltage	4.2	V
Nominal Voltage	3.6	V
Minimum Voltage	2.5	V
Max output current	45	A
Max continuous current	30	A
Max Cell temp	60	C
Cell Chemistry	LiPO4	
Datasheet	https://enepaq.com/v	
SDS	https://enepaq.com/v	
Additional Comments	Cells are assembled in Energus li1x6pvtc6t packs	

Conductors OK

Manufacturer	Part Number	Size [mm^2]	Voltage [V]	Ampacity [A]	Max Temp [C]	Datasheet	Location Used		Additional Comments	Rules References
LAPP	3806143	25	600	136	110	https://lappaustralia.	Accumulat	OK	Used between Accumulator and HIP	EV.5.2.1
LAPP	3806142	16	600	102	110	https://lappaustralia.	Accumulat	OK	Used for short distances in the Accumulator	EV.5.2.5
LAPP	16001303	4	750	23	110	https://lappaustralia.	Charger	OK	5 core length used between 3 phase outlet and charge	EV.6.3
ELECTRA	SRCV3025	2.5	750	21	110	https://cdn.shopify.c	Charger	OK	4 core, used within charger to connect power supplies	EV.6.3.2
ELECTRA	SRCV2025	2.5	750	22	110	https://cdn.shopify.c	Charger	OK	DC, used within charger to connect Phase to Phase DC	EV.6.6.4
HELUKABEL	29468	1.5	550	16	110	https://assets-cdn.he	Charger	OK	Used as Active, Neutral and Earth power outlet connec	
ELETECK	E254881	0.33	600	5	105	https://docs.rs-online	Accumulat	OK	Used as auxilary wire when nessecery for HV, Such as	
EXRAD	EXRAD-XH	50	1000	218	150	https://www.champca	HIP to Mot	OK	Supplied by Cascadia as per of the Motor Controller as	
EXRAD	EXRAD-XH	50	600	218	150	https://www.champca	Motor Con	OK	Supplied by Cascadia as per of the Motor Controller as	

Fuses OK

Manufacturer	Part Number	Voltage [V]	AC/DC	Current [A]	Interrupt [A]	Datasheet	Location Used		Additional Comments	Rules References
Littlefuse	L50QS100	500	DC	100	50,000	https://au.mouser.co	Top Plate	OK	Main fuse in accumulator	EV.5.2.1
Bel Fuse	C1Q 1	63	DC	1	50	https://www.belfuse.	BMS	OK	Voltage tap fuses mounted on CANaMon	EV.6.6
Energus	Internal	50	DC	270	500	https://enepaq.com/v	Accumulat	OK	Parallel fuses incorporated into each Energus block	
Bel Fuse	0ADKC400	500	DC	4	300	https://au.mouser.co	HIP	OK	DCDC HV Fuse	
Adler	APD05007	1000	DC	0.5	30,000	https://enertronic.es/	Accumulat	OK	Accumulator inline top plate fuses	

Resistors

OK

Manufacturer	Part Number	Voltage [V]	Resistance [Ω]	Power [W]	Power for 15 sec [W]	Heatsink [°C/W] (Enter "None" if no heatsink)	Datasheet	Location Used		Additional Comments & Heatsink Part Number(s)
Arcol	HS25 560I	550	560	25	75	None	https://docs.rs-online	Accumulat	OK	Precharge
Riedon	PF2472-3K	700	3300	100	500	19	https://riedon.com/m	HIP - Discl	OK	Discharge
Te Connectivit	SMF515KF	500	15000	5	5	None	https://www.te.com/	HIP	OK	Body Protection Resistors

Contactors/Relays

OK

Manufacturer	Part Number	Voltage [V]	Continuous Current [A]	Max Interrupt Current [A]	Normal State	Datasheet	Location Used		Additional Comments	Rules References
GigaVac	EPICGX	800	225	500	NO	https://www.mouser.	Accumulat	OK	Isolation Relays	EV.5.2.1
Omron	G7L-2A-X-	1000	20	20	NO	https://xonstorage.bl	Accumulat	OK	Precharge relay	EV.5.4
Cynergy	DBT71210	7000	2	3	NC	https://www.cynergy	HIP - Discl	OK	Discharge relay	EV.5.6

Connectors

OK

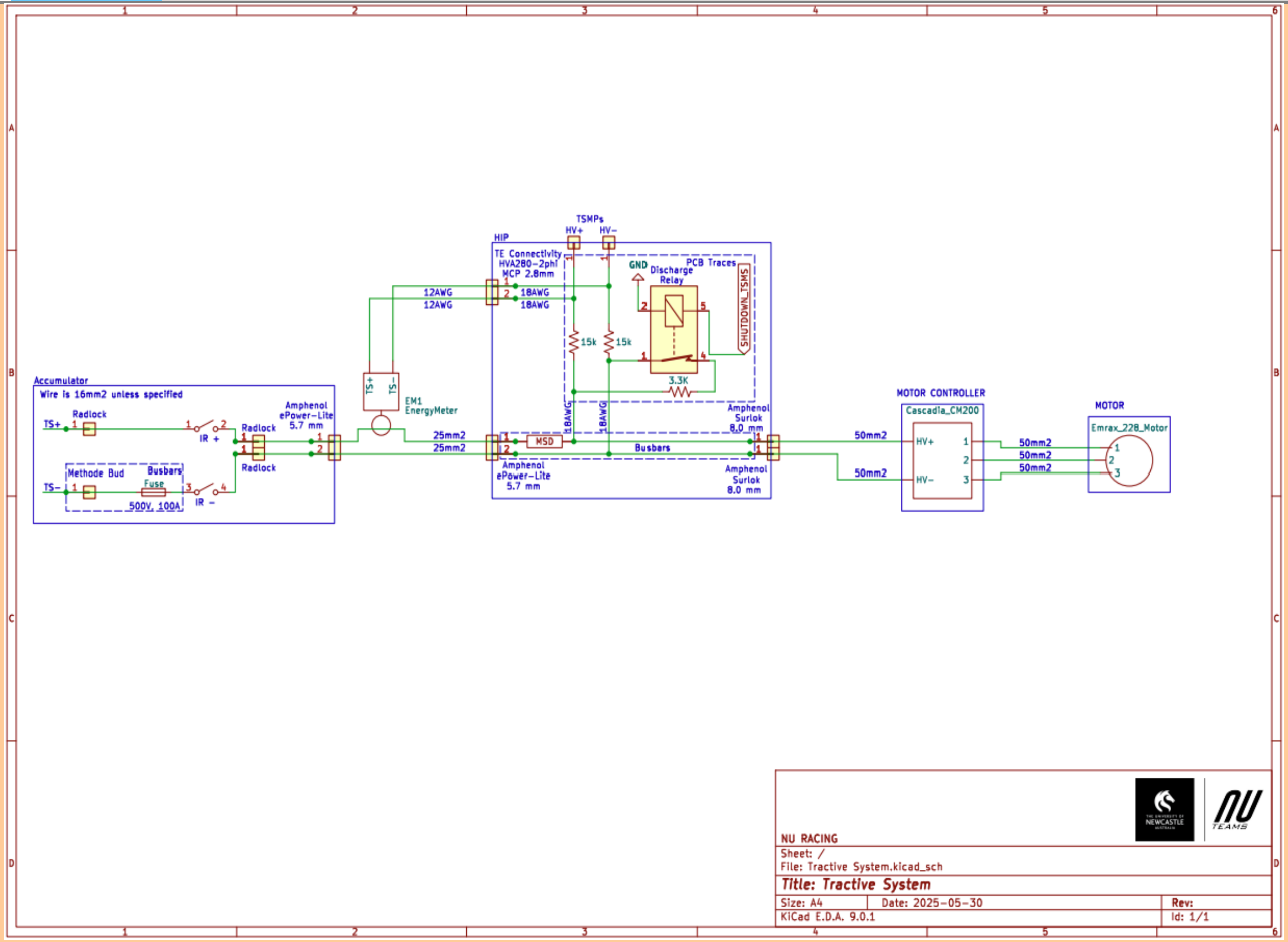
Manufacturer	Part Number	Voltage [V]	Ampacity [A]	Wire Size [mm^2]	Datasheet	Location Used		Additional Comments	Rules References
Amphenol	ELPA2A25	850	120	25	https://www.amphen	Accumulat	OK	HV loom from HIP to Accumulator, plug side	EV.5.2.1
Amphenol	ELRA2A03	850	120	25	https://www.amphen	Accumulat	OK	Located on the HIP and Accumulator lid, recepticle bus	EV.5.9
Methode	EBC 5.7mı	600	320	60	https://www.method	Accumulat	OK	Methode bud used on negative HV connection in accur	EV.6.3.3
Amphenol Tecl	RL00571-1	1000	120	25	https://au.mouser.co	Accumulat	OK	Inline radlok connectors, used on the positive HV conn	
Amphenol	SLPPB50B	1000	250	50	https://amphenol-ind	HIP - Moto	OK	HV loom from HIP to Motor Controller, plug side, 90 de	
Amphenol	SLPPB50B	1000	250	50	https://amphenol-ind	HIP - Moto	OK	HV loom from HIP to Motor Controller, plug side, 270 c	
Amphenol	SLPIRBBPS	1000	250	50	https://amphenol-ind	HIP Enclos	OK	Located on HIP, recepticle bus bar, 90 degree keyway	
Amphenol	SLPIRBBPS	1000	250	50	https://amphenol-ind	HIP Enclos	OK	Located on HIP, recepticle bus bar, 270 degree keywa	
Te Connectivit	4-2103177	600	27	12	https://www.te.com/	HIP - Ener	OK	Plug, connects DCDC and HIP HV connection. Current	
Te Connectivit	2103124-4	600	30	12	https://www.te.com/	HIP Enclos	OK	Recepticle, connects DCDC and HIP HV connection. Cu	
Hirose Electric	EM30MSD(	1500	200	40	https://www.hirose.c	MSD	OK	MSD	

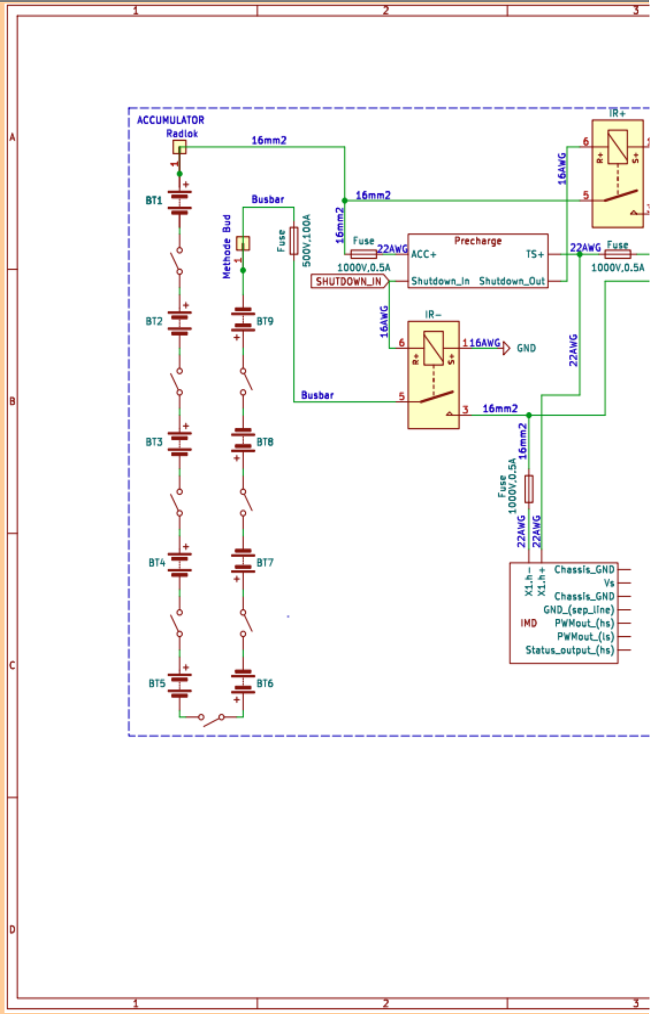
Insulating Materials									OK		
Manufacturer	Part Number	Thickness [mm]	Dielectric strength [V/mm]	Voltage Rating [V]	Max Temp [C]	Flammability Rating	Datasheet	Location Used		Additional Comments	Rules Re
ePOL	PEEK-NAT/	10	24000	**240000**	340	UL94 V-0	https://d347awuzx0k	Accumulat	OK	Accumulator top plate material	EV.5.2.1
RS PRO	20305179	1	31500	**31500**	130	UL94 V-0	https://docs.rs-online	Accumulat	OK	Accumulator wall lining material	EV.5.2.2.a
DuPont	Nomex 41(	0.25	33000	**8250**	260	UL94 V-0	https://www.swiftsup	Accumulat	OK	Accumulator lid lining material	EV.5.2.3
PCB	KB6165F T	1.6	25000	**40000**	170	UL94 V-1	https://odakpcb.com/	Accumualt	OK	Accumulator cell segment lining and cell PCB material	EV.5.4.4
											EV.6.2
											EV.6.5.5.a

Rules References
EV.5.2.1 EV.5.6 EV.5.8.4

ferences

Status	OK



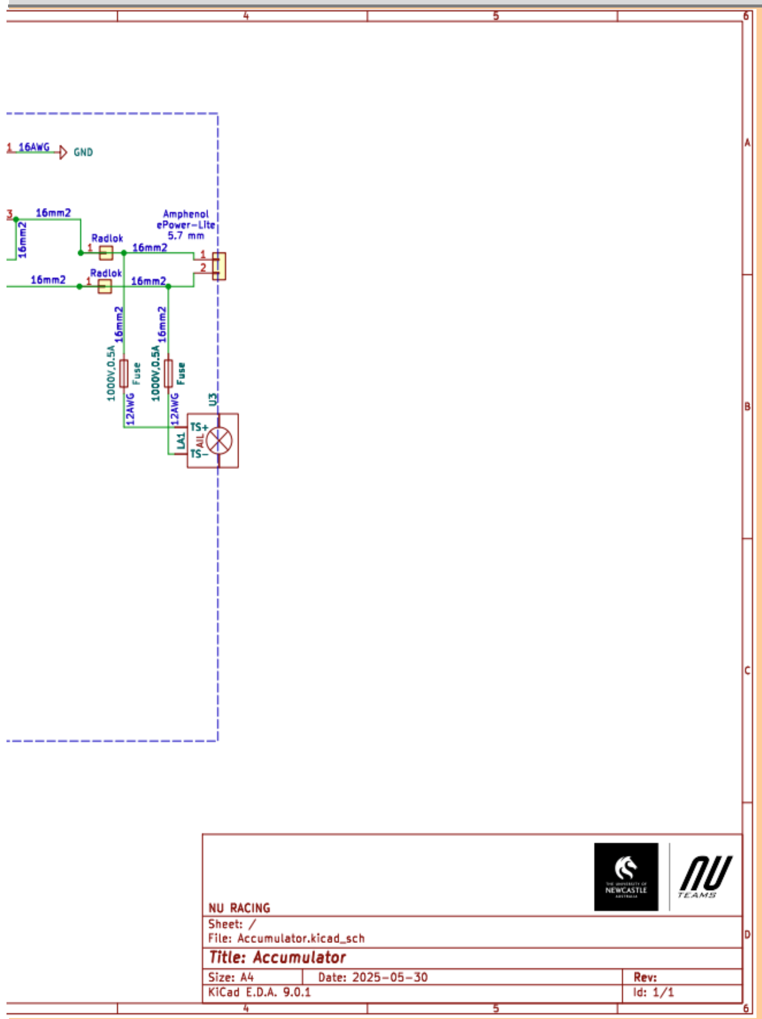


Rules References	Vehicle Tractive System Schematic Shows		OK
	Shows details of all TS circuits outside of accumulator	TRUE	OK
	Accumulator is shown as a single element (without internal details)	TRUE	OK
	All wire gauges labeled	TRUE	OK
	All fuses labeled with ampacity	TRUE	OK
EV.6.6.6	All wires > 150mm have a fuse within the first 150mm from the source	TRUE	OK
	Fuse locations represent physical location	TRUE	OK
	Enclosures shown	TRUE	OK
	Connectors labeled with Make/Model	TRUE	OK
	Standard schematic symbols used	TRUE	OK
	All text is readable (zooming is allowed)	TRUE	OK
EV.5.8	TSMPs	TRUE	OK
	Motor Controller	TRUE	OK
	Motor	TRUE	OK
EV.5.8.5	TSMPs not fused	TRUE	OK

Rules References	Must be shown on at least one schematic		OK
EV.7.6	IMD	TRUE	OK
EV.5.5	MSD	TRUE	OK
EV.5.6	Precharge	TRUE	OK

Rules References	Accumulator TS S
EV.6.6.6	All wires > 150mm have a f
EV.5.4	
EV.5.3	
	Cells (details of
EV.5.7	Accumulator

EV.5.6	Discharge	TRUE	OK	
EV.3.2; EV.5.1.3	Energy Meter (shunt and TS sense connection)	TRUE	OK	
EV.5.6.3.b	Discharge still functions when MSD is opened	TRUE	OK	
EV.5.6.3.c	Discharge is not fused	TRUE	OK	



Schematic Shows		OK
Details of all TS circuits within accumulator	TRUE	OK
All wire gauges labeled	TRUE	OK
All fuses labeled with ampacity	TRUE	OK
use within the first 150mm from the source	TRUE	OK
Fuse locations represent physical location	TRUE	OK
Connectors labeled with Make/Model	TRUE	OK
Standard schematic symbols used	TRUE	OK
All text is readable (zooming is allowed)	TRUE	OK
IRs	TRUE	OK
Maintenance plugs	TRUE	OK
all intercell connections may be abstracted)	TRUE	OK
Indicator (must be powered directly by TS)	TRUE	OK

Additional Comments

Each of the 9 accumulator segments are comprised of 12 li1x6pvtc6t packs arranged in series. These li1x6pvtc6t packs consist of 6 inidivdual Sony Murata 18650 cells arranged in parallel.

Status	OK		
Segment-to-Segment Connection	Series		OK
Maximum TS Voltage	453.6	V	OK
Nominal TS Voltage	388.8	V	
Total Cells	648		

Segment Data

OK

Cell Connection Order	Parallel then Series						OK
Segment	Parallel cells	Series cells	Temp Sensors	Accumulator Number	Max Voltage [V]	Segment Energy [MJ]	
1	6	12	20	1	50.4	3.374784	OK
2	6	12	20	1	50.4	3.374784	OK
3	6	12	20	1	50.4	3.374784	OK
4	6	12	20	1	50.4	3.374784	OK
5	6	12	20	1	50.4	3.374784	OK
6	6	12	20	1	50.4	3.374784	OK
7	6	12	20	1	50.4	3.374784	OK
8	6	12	20	1	50.4	3.374784	OK
9	6	12	20	1	50.4	3.374784	OK
10					0	0	
11					0	0	
12					0	0	
13					0	0	
14					0	0	
15					0	0	
16					0	0	
17					0	0	
18					0	0	

BMS

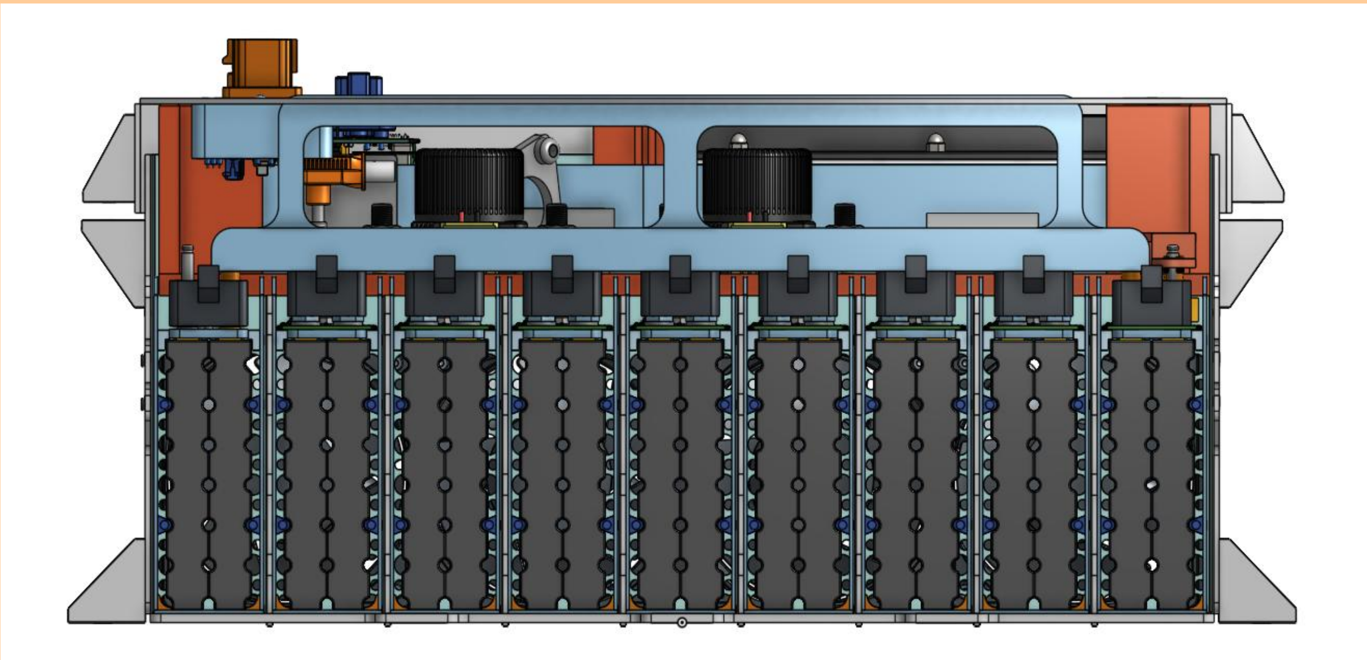
BMS Type	Centralized	
BMS Source	Purchased	
Make	Ewert Energy	
Model	Orion 2	
Galvanic Isolation TS to GLV	1500	V
Galvanic Isolation between segments	2500	V

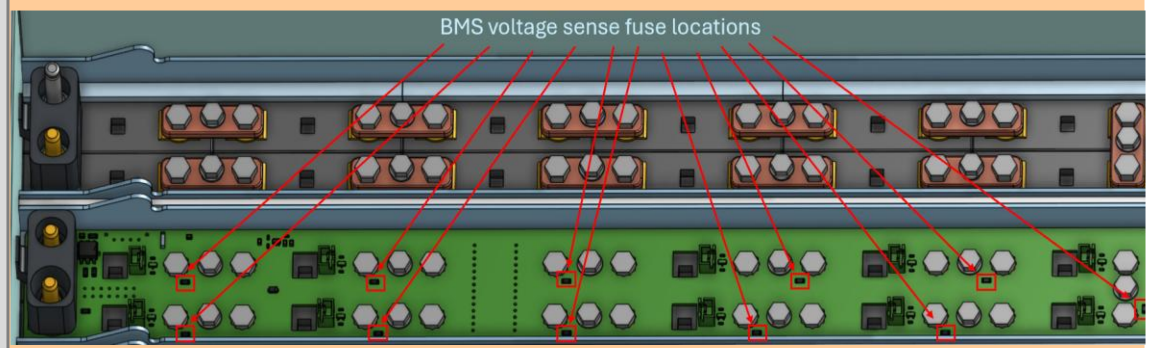
Voltage Sense Leads

Maximum Unfused Lead Length	16	mm
Sense Lead Overcurrent Protection Device	Bel Fuse C1Q 1	
Overcurrent Location	Other	
Overcurrent Other Location	CANaMon	
Overcurrent Voltage Rating	63	V

Temperature Sensors

Temp Sensor Location	On cell terminal	
Cells sensed per sensor	2	cells
Thermal Contact material used		
Thermal contact material datasheet		
Thermal contact material conductivity		W/(m·K)
Thermal contact material thickness		mm
Percent of cells sensed	55.5556	%





Maintenance Plug(s) Picture Shows		OK
Unique Configuration of plugs	TRUE	OK
Non-conductive on un-necessary surface	TRUE	OK
Accessible without any additional accumulator disassembly	TRUE	OK
Positive Locking	TRUE	OK

Additional Comments	Rules References
Busbar positively locked by accumulator lid.	EV.5.3

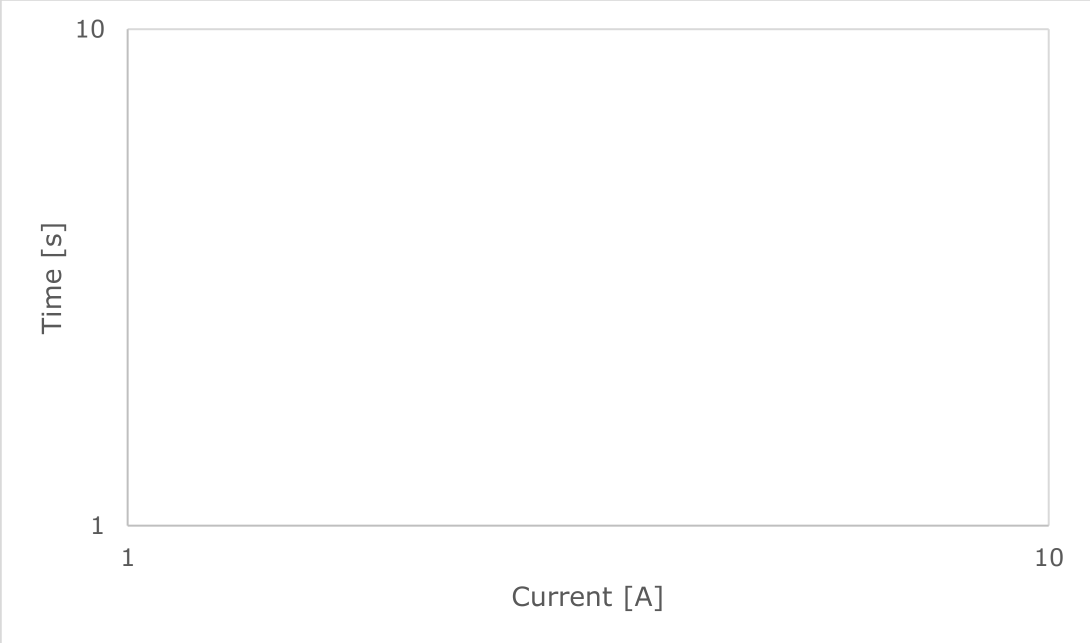
Cell Connections Picture Shows	
Bolted connections have positive locking	TRUE
Location of parallel cell overcurrent for parallel cells	TRUE
BMS voltage sense fuse location	TRUE

Additional Comments

Parallel Cell Fusing			OK	Rules Referenes
Source	Purchased		OK	EV.6.6.3
Make/Model	Energus Internal		OK	EV.6.6.5
Continuous Current	270	A	OK	

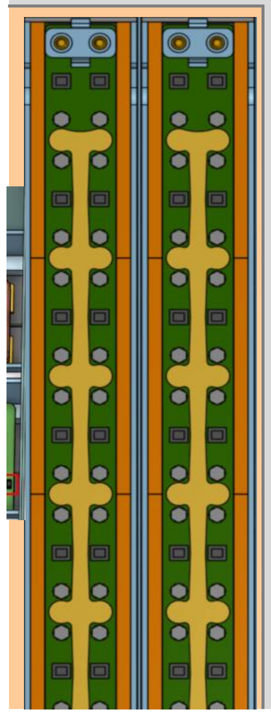
Test Data Summary	
Tests at current 1	0
Tests at current 2	0
Tests at current 3	0
Tests at current 4	0

See Fusible Link Testing Guideline on FSAEOnline.com



Testing
Test 1
Test 2
Test 3
Test 4
Test 5
Test 6
Test 7
Test 8
Test 9
Test 10
Test 11
Test 12
Test 13
Test 14
Test 15
Test 16

OK	**Maintenance Plugs**			OK	**Insulating Materials**			OK	
OK	Connector	Methode EBC 5.7mm		OK	Between cells and accum. container	RS PRO 20305179		OK	Due to design of temp
OK	Ampacity	320	A	OK	Voltage Rating	31500	V	OK	segment will not have
OK					Temperature Rating	130	C	OK	
OK					Flammability Rating	UL94 V-0			
OK	**IR**				On top of cells	PCB KB6165F TG 149		OK	
	Make/Model	GigaVac EPICGX		OK	Voltage Rating	40000	V	OK	
	Contact Voltage Rating	800	V	OK	Temperature Rating	170	C	OK	
	Max switching current	500	A		Flammability Rating	UL94 V-1			
OK	Max continuous current	225	A	OK	Separating IRs and Fuse from cells	ePOL PEEK-NAT/BLK		OK	
OK	**Main Fuse**				Voltage Rating	240000	V	OK	
OK	Main Fuse	Littlefuse L50QS100.T		OK	Temperature Rating	340	C	OK	
OK	Main fuse ampacity	100	A		Flammability Rating	UL94 V-0			
OK	Main fuse voltage	500	V	OK					
	MSD								
OK	MSD	Electric Group EM30M		OK					
	Ampacity	200	A	OK					
OK									
OK									



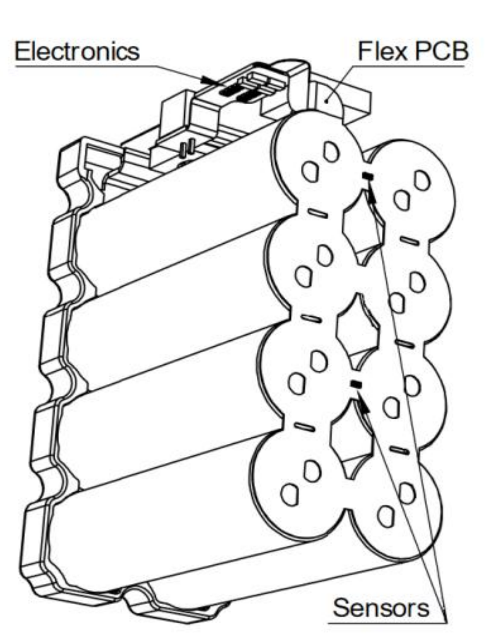
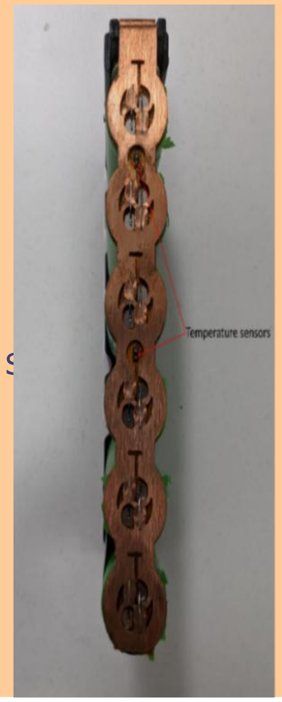
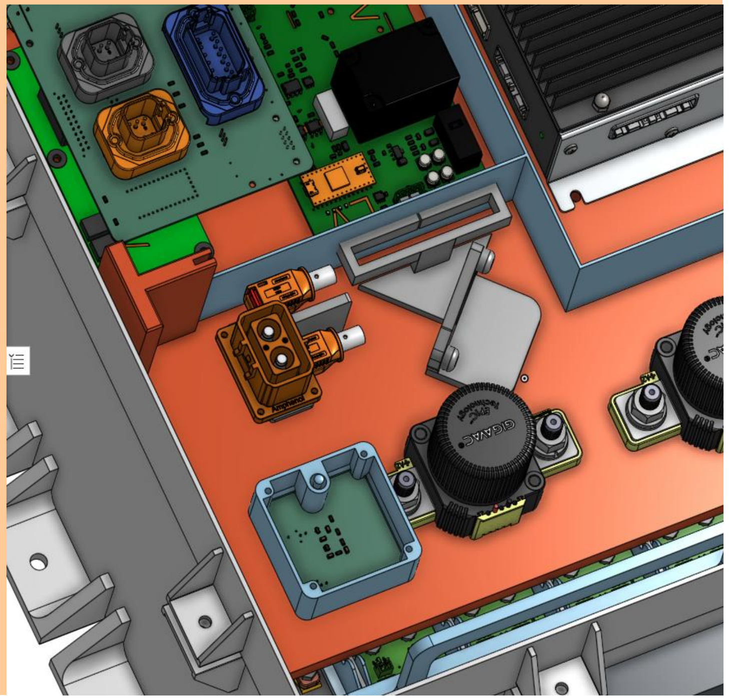


Figure 3: Sensor layout.





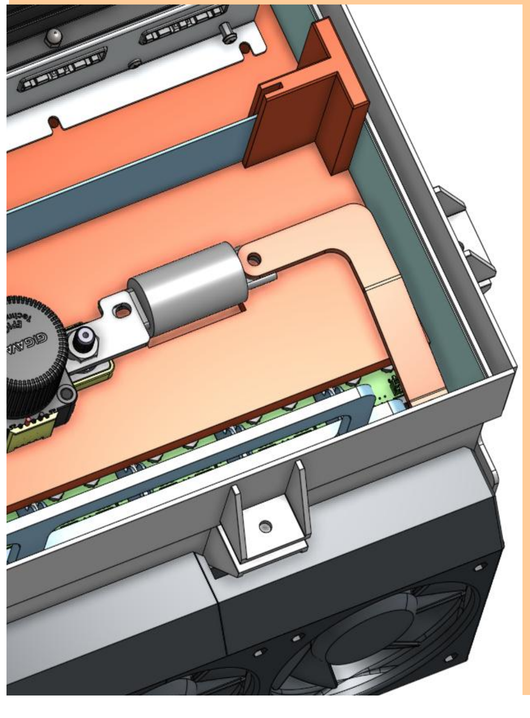
OK
OK
OK
OK
Rules References
EV.6.4.1 EV.6.4.3 EV.7.4.4.a

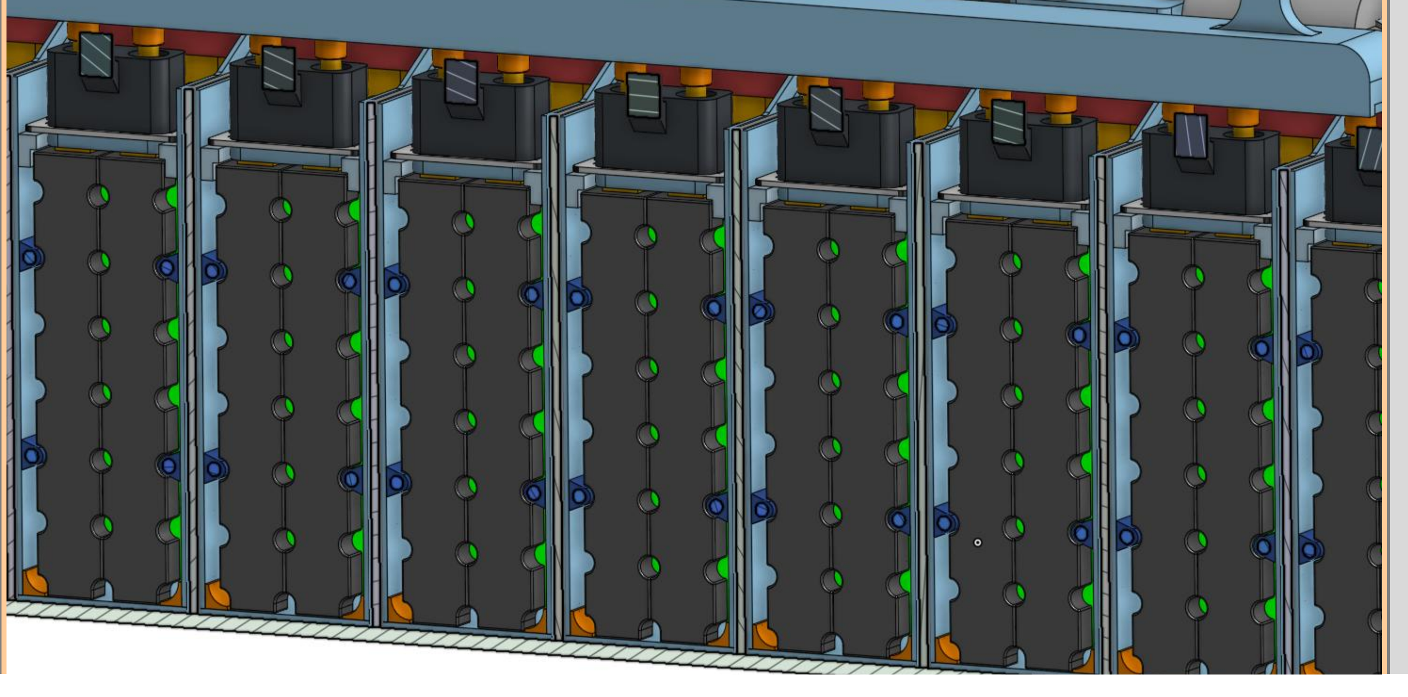
Temperature Sensor Picture Shows		OK
Temp sensor in direct contact with terminal or busbar	TRUE	OK
Distance from sensor to cell negative terminal connection	TRUE	OK
Additional Comments		Rules References
Please note that the li1x6pvtc6t packs used in our accumulator are configured in a 1x6 pack, not a 4x2 as pictured above. This image was taken from the pack's data sheet which covers a range of battery configurations. The locations of the sensors is the same for both		EV.7.5

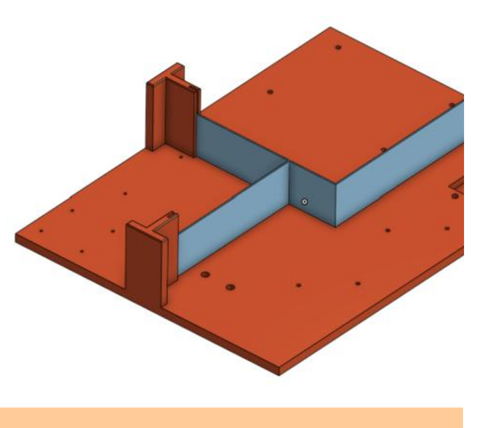
Fuse and IRs Separation Picture Sl
Fuse and IRs are separated
Additional Comments

Additional Comments

erature monitoring PCB the centre 2 battery packs in each
temperature monitored.



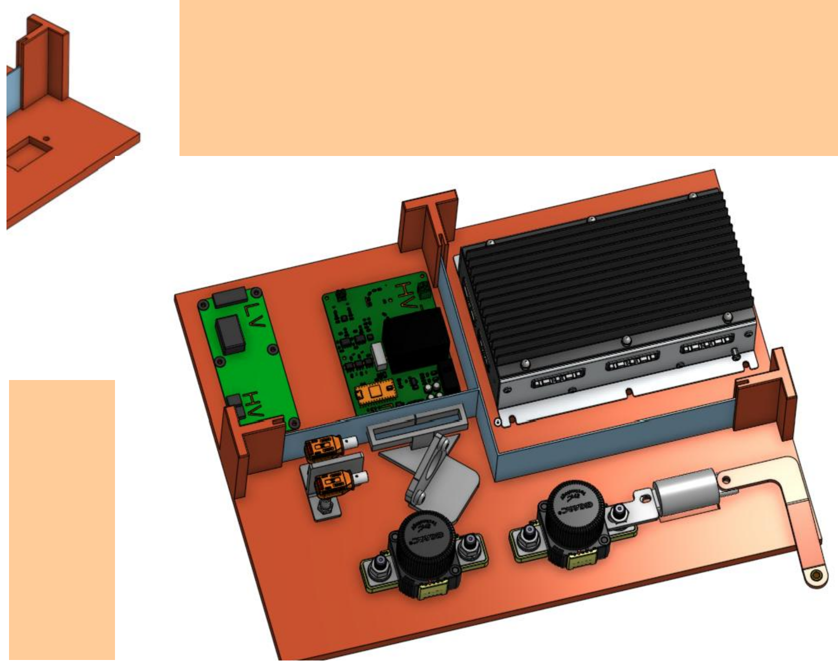




hows		OK
from cells	TRUE	OK
		Rules References
		EV.5.4.4

Segment Insulation Picture Shows		OK
Insulation between cells and accumulator container	TRUE	OK
Additional Comments		Rules References
		EV.5.2.2.a EV.5.2.3

Segment Ins
Insulation or
Additi


LV
HV

sulation Picture Shows		OK
ı top of cells (air not acceptable)	TRUE	OK
onal Comments		Rules References
		EV.5.2.2.a EV.5.2.3

Status	OK

Precharge — OK

Resistor	Arcol HS25 560R J		OK
# parallel	1		OK
# series	1		OK
Resistance	560	Ohms	
Cont power	25	W	
power @15 sec	75	W	
Voltage	550	V	OK
Bus Capacitance	0.65	mF	OK
End of precharge	Ratio		OK
ending ratio	90	%	OK
Peak power	367.416	W	
Peak current	0.81	A	
time to 90%	0.84	sec	
Precharge Res Energy to 90%	66.20	J	
Average Power	79.0	W	
Relay	Omron G7L-2A-X-L		OK
Relay voltage rating	1000	V	OK
Relay current rating	20	A	OK

Discharge — OK

Resistor	Riedon PF2472-3K3F1		OK
# parallel	1		OK
# series	1		OK
Resistance	3300	Ohms	
Cont power	100	W	
power @15 sec	500	W	OK
Voltage	700	V	OK
Peak power	62.3494	W	
Peak current	0.14	A	
Discharge time (to 60V)	4.33906	s	OK
Relay	Cynergy DBT71210		OK
Relay voltage rating	7000	V	OK
Relay current rating	2	A	OK
Normal State	NC		OK

Rules References
EV.5.6; EV.7.2.2.c

Vehicle TSMP

Resistor	Te Connectivity SMF5	
# parallel	1	
# series	1	
Resistance	15000	Ohms
Cont power	5	W
Voltage	500	V

Charger TSMP

Same parts as vehicle TSMP?	Yes

Additional Comments

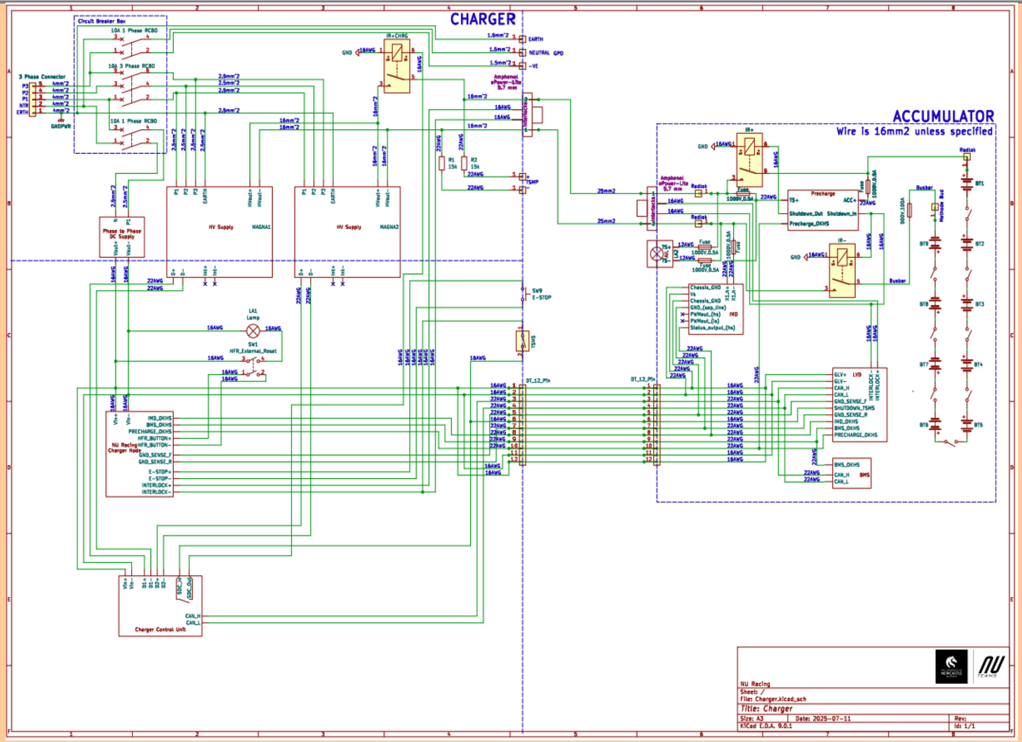
OK	Rules References
OK	EV.5.8.4
OK	
OK	
OK	
OK	
OK	

OK	Rules References
OK	EV.5.8.4

Status	OK

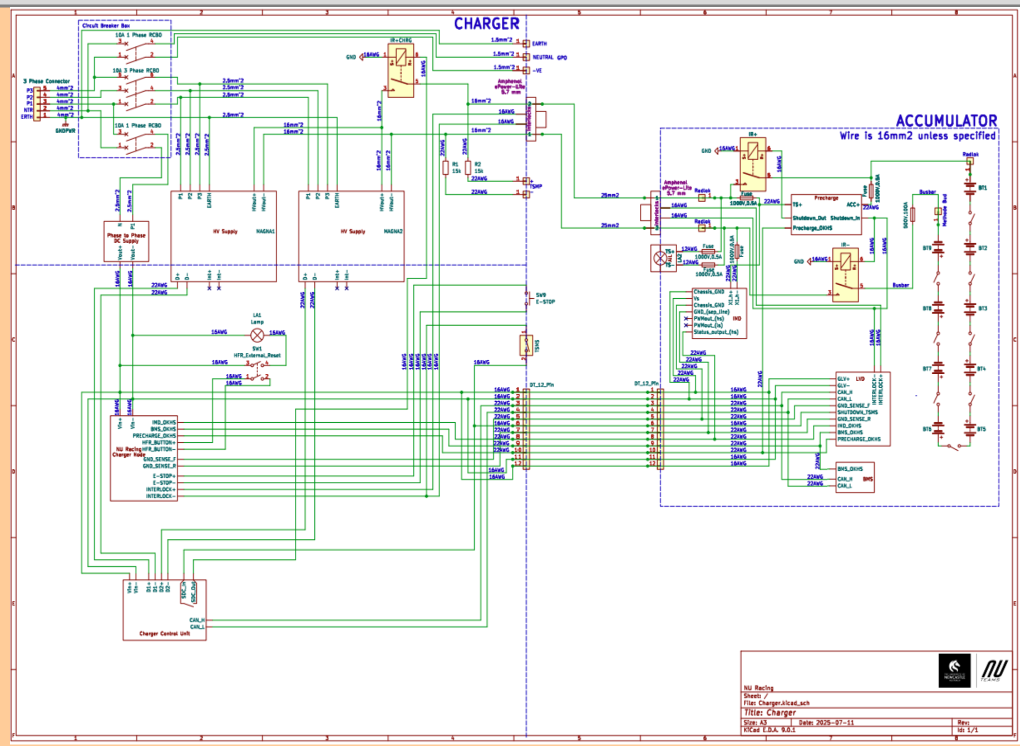
Charger			OK
Make	Magna Power		OK
Model	SL series		OK
Datasheet	https://magna-power		OK
Input Voltage	380/400	V	OK
Input Current	11	A	OK
Output Voltage	500	V	OK
Output Current	10	A	OK
Power	4	kW	OK
Input to Output isolation	2.5		OK

Additional Comments



Charging Tractive System Schematic Shows			OK
Rules References			
EV.6.6	Fuse	TRUE	OK
	Connectors	TRUE	OK
	Wire Gauge	TRUE	OK
EV.8.4	IMD	TRUE	OK
EV.8.4.2	IRs	TRUE	OK
EV.8.2.6	TSMPs	TRUE	OK

Additional Comments



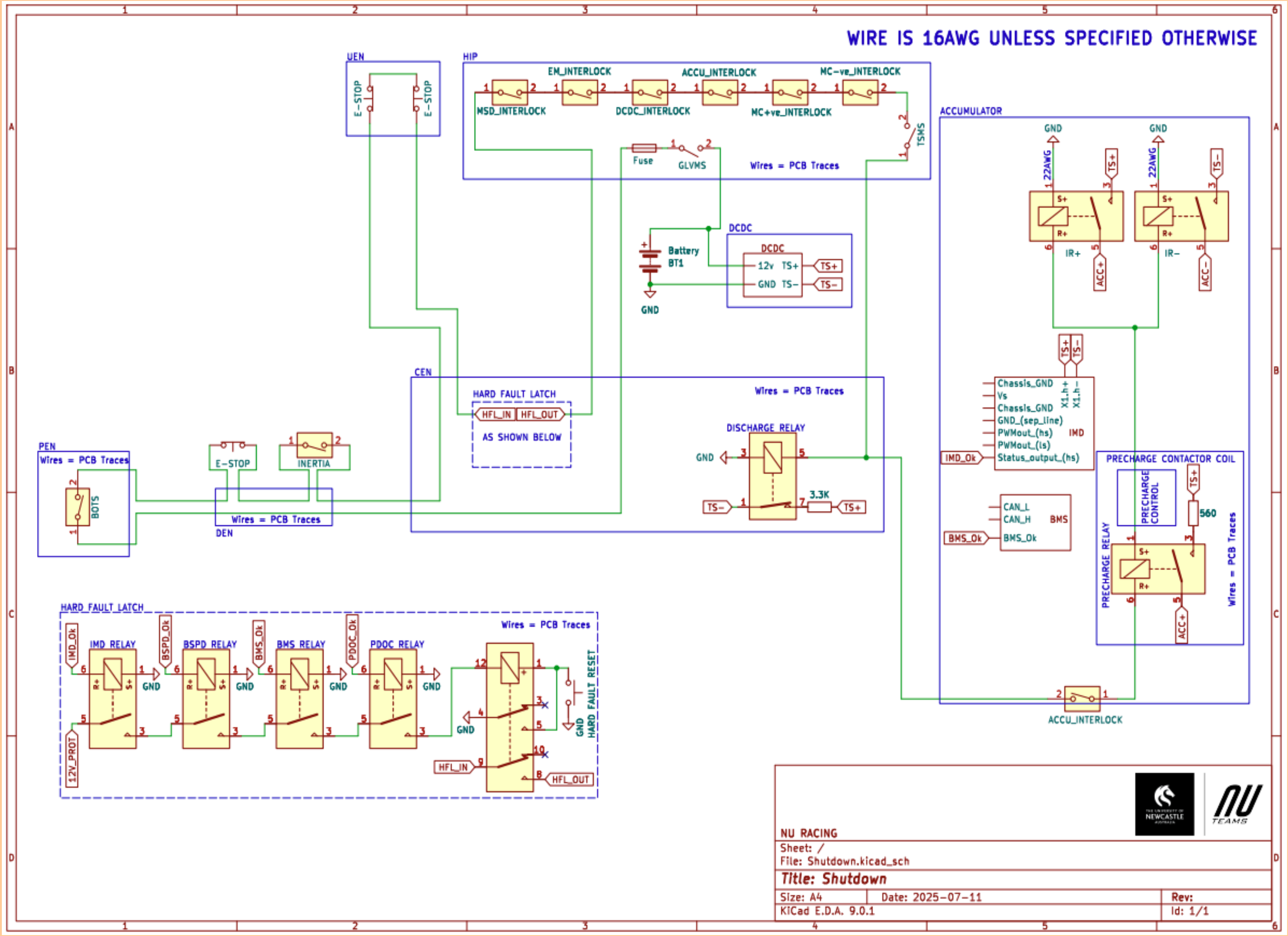
Charging Shutdown Circuit Shows			OK
Rules References			
EV.8.3	IMD Control and Powerstage	TRUE	OK
EV.8.3	AMS	TRUE	OK
EV.8.2.7	Charger Shutdown Button	TRUE	OK
EV.8.4	IRs	TRUE	OK
EV.5.9	Charge Connector Interlock	TRUE	OK
EV.8.2.4; EV.8.4.2.c	Charge Control (CAN, Enable, etc.)	TRUE	OK

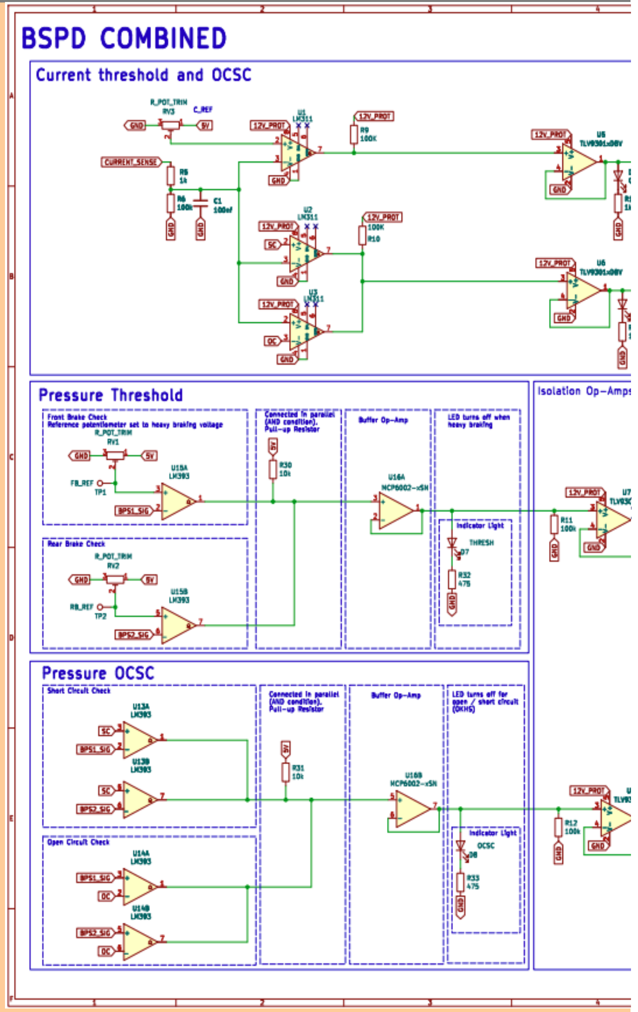
Additional Comments

Status	BLANK

IMD			OK
Rules Reference	EV.7.6		
Make/Model	Bender IR155-3204		OK
Setpoint	250	kΩ	OK

BSPD			OK
Rules Reference	EV.7.7		
Current Sensor Make/Model	Tamura L31S100S05F		OK
Current Sensor Datasheet	https://au.mouser.co		OK
Trip Current	12.5	A	OK
Sensor Output @ Trip Current	2.57	V	OK
Accuracy @ Trip Current	0.1	A	OK



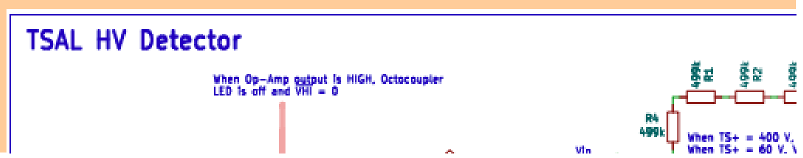


Shutdown Circuit Schematic Shows			OK
Rules References			
EV.7.9.2	GLVMS	TRUE	OK
EV.7.9.3	TSMS	TRUE	OK
EV.7.10.3	2 Side Shutdown Buttons	TRUE	OK
EV.7.10.4	1 Cockpit Shutdown Button	TRUE	OK
T.3.3	Brake Over Travel Switch (BOTS)	TRUE	OK
T.9.4	Inertia Switch	TRUE	OK
EV.7.3	BMS	TRUE	OK
EV.7.6	IMD	TRUE	OK
EV.7.7	BSPD	TRUE	OK
EV.5.5.3	MSD Interlock	TRUE	OK
EV.7.1.4	IMD, AMS, BSPD have independent power stages	TRUE	OK
EV.4.1.3.a	Outboard Wheel Motor Interlocks (if applicable)	TRUE	OK
	GLV Battery	TRUE	OK

BSPD System Sc

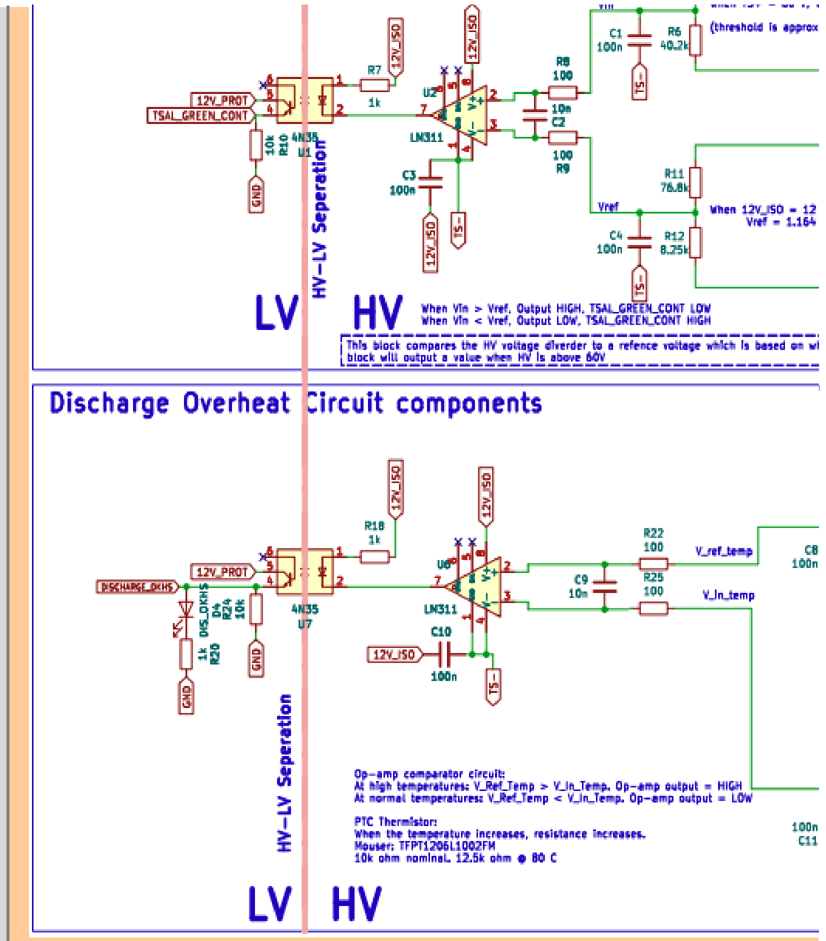
BSPD Curren
BSPD Brake

Additional Comi

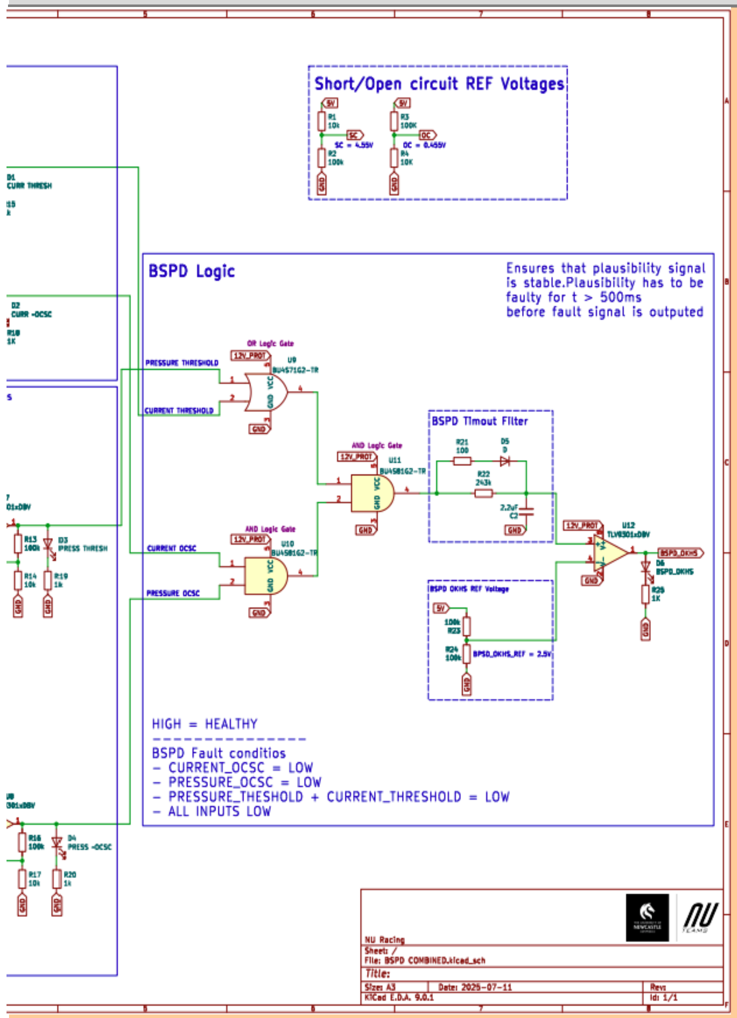


Public

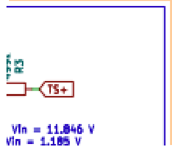
	HV DC/DC converter GLV connection (if applicable)	TRUE	OK
T.9.2.2	GLV Fuse	TRUE	OK
	Wire Gauge	TRUE	OK
EV.7.1.2; EV.7.1.5	Precharge Contactor coil	TRUE	OK
EV.7.1.2; EV.7.1.5	Positive IR coil	TRUE	OK
EV.7.1.2; EV.7.1.5	Negative IR coil	TRUE	OK
EV.5.6.3.a	Discharge Contactor coil	TRUE	OK
EV.7.9.3.b	TSMS is last switch	TRUE	OK
EV.7.9.2.a	GLVMS is first switch	TRUE	OK
	Uses proper electrical schematic symbols	TRUE	OK
	Labels/text are readable (zooming in is acceptable)	TRUE	OK
EV.5.9	Interlocks on TS connectors outside of an enclosure	TRUE	OK
Additional Comments		Rules References	
		EV.7.1 EV.7.2	

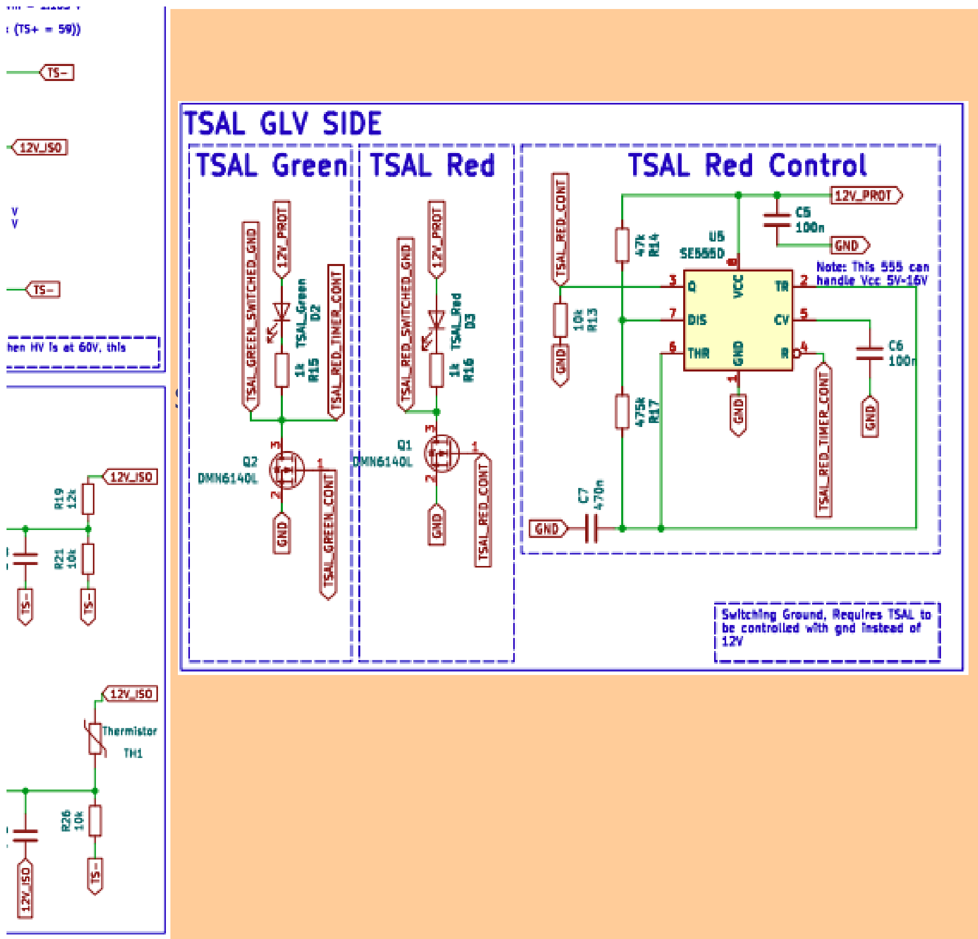


Ready to Move Ligh	
Light controlled by >60V presen	
Light	
DOT I	
Additional Comm	



chematic Shows		OK
t Sensor input open/short circuit protection	TRUE	OK
Position input open/short circuit protection	TRUE	OK
Shutdown circuit powerstage	TRUE	OK
ments	Rules References	
	EV.7.7	

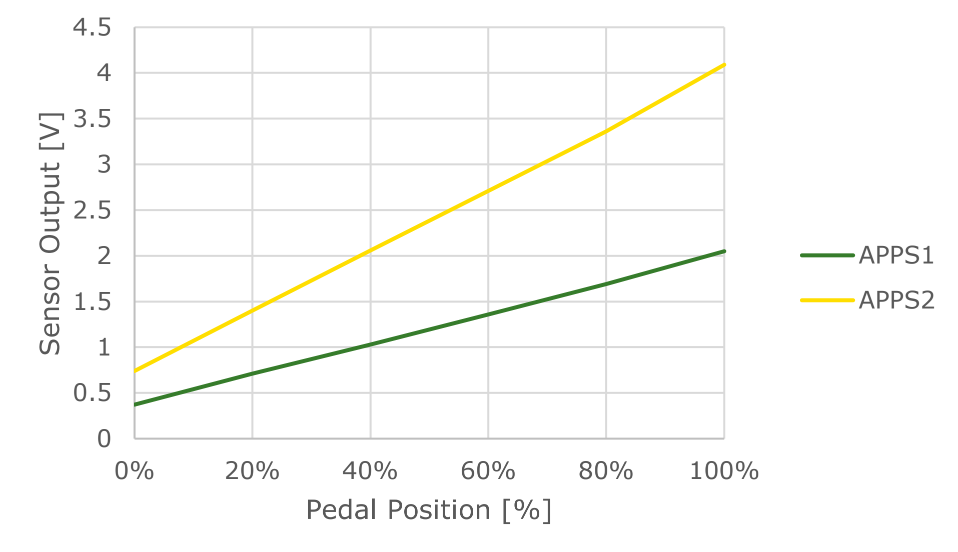




t Schematic Shows		BLANK
t in TS (hardware-based voltage detection)	TRUE	OK
Light powered by GLV	TRUE	OK
flashes at 2–5 Hz when >60V present in TS	TRUE	OK
Light Source	Team Built	OK
FMVSS 108 compliant		BLANK
ments	Rules References	
	EV.5.10	

Public

Status	OK		
APPS Sensor Transfer function			OK
Rules Reference	T.4.2.3		
Position	APPS1 Output [V]	APPS2 Output [V]	
0%	0.372	0.74	OK
20%	0.711	1.4	OK
40%	1.03	2.06	OK
60%	1.36	2.71	OK
80%	1.69	3.36	OK
100%	2.05	4.09	OK



Torque Control Path Security Checks

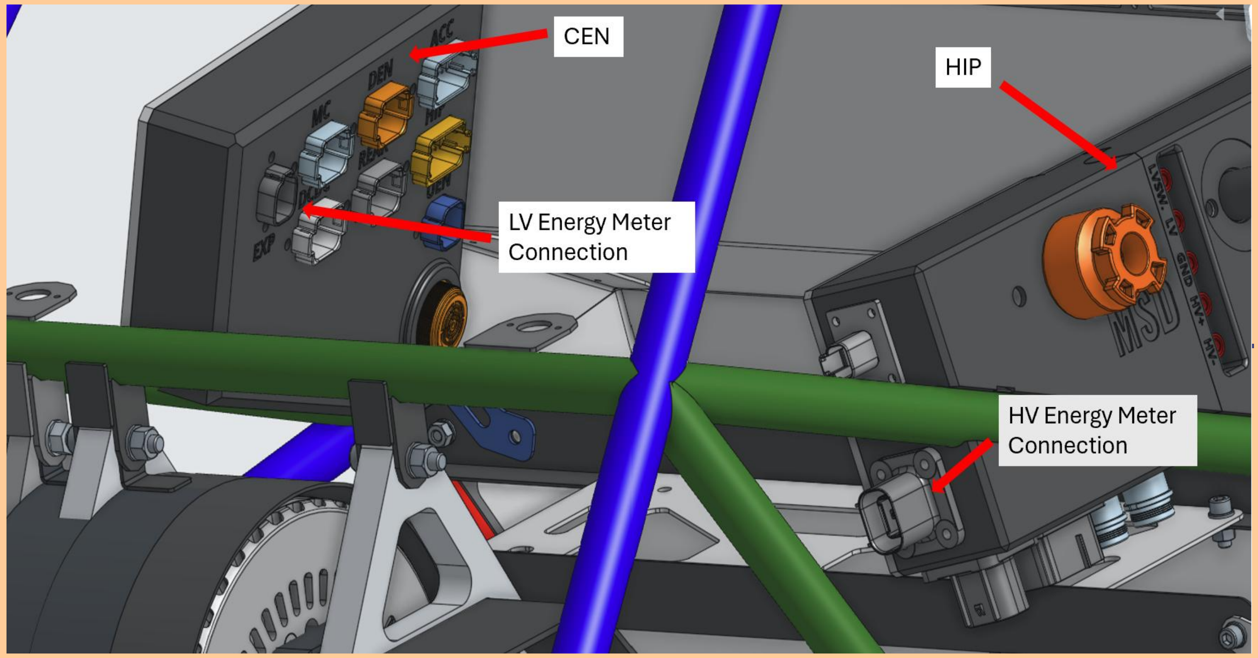
Source Device	Destination Device	Communication	Redundant	Out Of Range	Correlation	Checksum	Timeout
APPS	Pedal Box ECU	Analog	2 Sensors	Yes	No	No	No
Pedal Box ECU	DEN	Digital	No	No	No	CAN	Yes
DEN	CEN	Digital	No	No	No	CAN	Yes
CEN	Motor Controller	Digital	No	No	No	CAN	Yes
Motor Controller							

	OK	
Other		Additional Comments
	OK	
	OK	
	OK	
	OK	

Status	BLANK		

Motor Controller			OK
Make	Cascadia		OK
Model	CM200DX		OK
Datasheet	https://www.cascadia		OK
Galvanic Isolation between TS and control	1500	V	OK

Motor		OK
Make	Emrax	OK
Model	228 MV	OK
Datasheet	https://emrax.com/w	OK



.2

Picture of Tractive System Status Indicator on car to prove compliance with EV.5.11

Energy Meter Download Picture Shows		OK
Energy Meter Download Connector	TRUE	OK

Additional Comments

Tractive System Status Indicator Shows		BLANK
Tractive System Status Indicator placement in vehicle		BLANK

Additional Comments